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Q1a. The Diffie-Hellman key exchange scheme can secure the eavesdropper attacks. The eavesdropper to monitor the public communication channel can get the public key of two people. But the eavesdropper hard to use the public to derive the private keys based on the discrete logarithm problem. But the Diffie-Hellman key exchange scheme cannot defend Man-in-the-Middle Attack. Because Diffie-Hellman did not authenticate the exchange person, the third person can intercept and masquerade as the exchange person.

Q1b. Because the ElGamal encryption scheme cannot defend Known-plaintext attack. If attacker knows one plaintext, he can easily recover the other. Because the structure of two different ciphertexts only changes the plaintext and the random number k . If k does not change and the attacker knows one plaintext, he can easily recover the other plaintext. Even those attackers do not know k .

Q1c.

Maybe can not defend Man-in-the-Middle Attack. Because this method did not provide an authentication method. An attacker can change the other public key in the blackboard to their key. Or impersonate another to write his key with another person's name. So that the attacker can intercept and decrypt their messages.

Q2a.

Only those blocks that are before the tenth plaintext block(one to nine) will be correct. So only 9 blocks of plaintext will be correct. Because CBC mode, the encrypted result in one block will be used in the next block. So the one bit of the tenth plaintext block is modified, it makes the encrypted result of the tenth block is wrong. And the wrong encrypted result of the tenth block will be used to encrypt the eleventh block, so the eleventh block will also be wrong. And keep going wrong to the last.

Q2b.

Because the tenth ciphertext block is changed after encryption, by CBC mode, the encrypted result in one block will be used in the next block encryption. If the change happens after all block encryption, it only change the tenth block and eleventh block. Because only the eleventh block decryption need to use the tenth block. All other 98 block are correct.

Q3a.

Because the location is limited to the set A to Z and knows that it uses a 24-hour format. And the plaintext and ciphertext structures are very simple. If we know their encryption scheme and some encryption method, make the same plaintext will always produce the same ciphertext. we can use the Brute Force Attack. The range of plaintext and ciphertext is too small. Also the Bob responds, only Yes and no are easier to use Brute Force Attack. Especially if the same date are made multiple times.

Q3b.

To add an unused word to increase the plaintext. Because some encryption methods, make the same plaintext always produce the same ciphertext. Just use Yes and No to be ans is easy let attacker to Brute Force Attack. Add some words with Yes and No make the ciphertext change.

Q4a.

In RSA, the message m must be less than n . While n is large (more than 1024 bits), the chosen exponent can lead to potential vulnerabilities due to a small message size. Alice can use padding to insert randomness into the message. To increase the M size.

Q4b.

$(e_A = 3, n)$ and $(e_B = 5, n)$

$C_A = m^{e_A} \bmod n = m^3 \bmod n$

$C_B = m^{e_B} \bmod n = m^5 \bmod n$

Use Extended Euclidean algorithm: $e_A = 3, e_B = 5$

$$\gcd(3,5)=1$$

$$3x+5y=1$$

$$m^{(3x+5y)}=m^1=m \pmod{n}=m$$

Eve can undo the encryption through the relationship $3x+5y=1$

Q4c.

$(e_1 = 3, N_1), (e_2 = 3, N_2), (e_3 = 3, N_3)$, respectively.

Assume that N_1, N_2, N_3

$$c_1=m^3 \pmod{N_1}$$

$$c_2=m^3 \pmod{N_2}$$

$$c_3=m^3 \pmod{N_3}$$

By Chinese Remainder Theorem

$$x \equiv c_1 \pmod{N_1}$$

$$x \equiv c_2 \pmod{N_2}$$

$$x \equiv c_3 \pmod{N_3}$$

So that, $x=m^3 \pmod{(N_1 * N_2 * N_3)}$

$m=\text{cube root of } x$

Q5a.

$$H(x \oplus y)=H(x) \oplus H(y)$$

$$H(0^k \oplus 0^k) = H(0^k) \oplus H(0^k)$$

$$H(0^k) = H(0^k) \oplus H(0^k).$$

$$\text{Because: } 0^k \oplus 0^k = 0^k$$

$$H(0^k) \oplus H(0^k) = 0^n$$

$$\text{So, } H(0^k) = 0^n$$

Q5b

A cryptographic hash function is strong collision-resistant if it is computationally infeasible to find any two distinct inputs x and y such that: $H(x) = H(y)$.

And this hash function: $H(x \oplus y) = H(x) \oplus H(y)$.

Assume x and x' , and $x' = x \oplus y$

$$H(x') = H(x) \oplus H(y).$$

And prove not strong-collision resistant, we need $H(x) = H(x')$.

And base on $H(x) = H(x) \oplus H(y) = H(x')$

$$H(y) = 0^n$$

So H is not strong-collision resistant.